

Default and Credit Rating Report

1. Loading data and cleaning

I first inspect the raw data to assess how and what to clean. The dataset has 145 columns, with many containing a large amount of missing values. The columns relevant to the current analysis, such as loan status, grade, and home ownership are all relatively complete.

To clean the data, I select only the relevant columns. Furthermore, I create a binary outcome variable for whether the loan status is fully paid off or not. The summary statistics of the cleaned data indicate outliers are present in the data. For instance, annual income has a maximum of 9 500 000. Similarly, debt-to-income also has a handful of extreme outliers, with a handful of pre-cleaning observations with a dti equal to 999. The cleaning drops the most extreme dti observations.

Table 1.1: Summary statistics

variable	mean	median	sd	min	max
emp_length	6.1	6	3.6	1.0	10
dti	18.8	18	10.5	0.0	822
annual_inc	78865.8	66000	80722.6	0.0	9550000
int_rate	13.0	12	5.0	5.3	31
default	0.2	0	0.4	0.0	1

2. Default rates on short-term loans

First, I filter to short-term loans, and then create a function which calculates the default rate for any group. Among short term loans (36 months), the default rate falls as home ownership improves. Renters default at 23.1%, compared to 19.9% for owners and 15.2% for those still paying a mortgage. This provides an indication that owning vs renting matters. Although, the fact that mortgage holders default less than owners is a less intuitive result.

Table 2.1: Default rate by home ownership

home_ownership	n	default_rate
MORTGAGE	139189	15.2
RENT	120789	23.1
OWN	37406	19.9
ANY	63	20.6

Table 2.2: Default rate by employment length

emp_length	n	default_rate
5	18329	18.8
10	96678	17.0
3	24559	19.3
4	17909	19.0
1	44381	19.6
8	12735	18.2
NA	21655	27.9
6	12526	17.9
9	10905	18.6
2	27805	19.0
7	9965	18.5

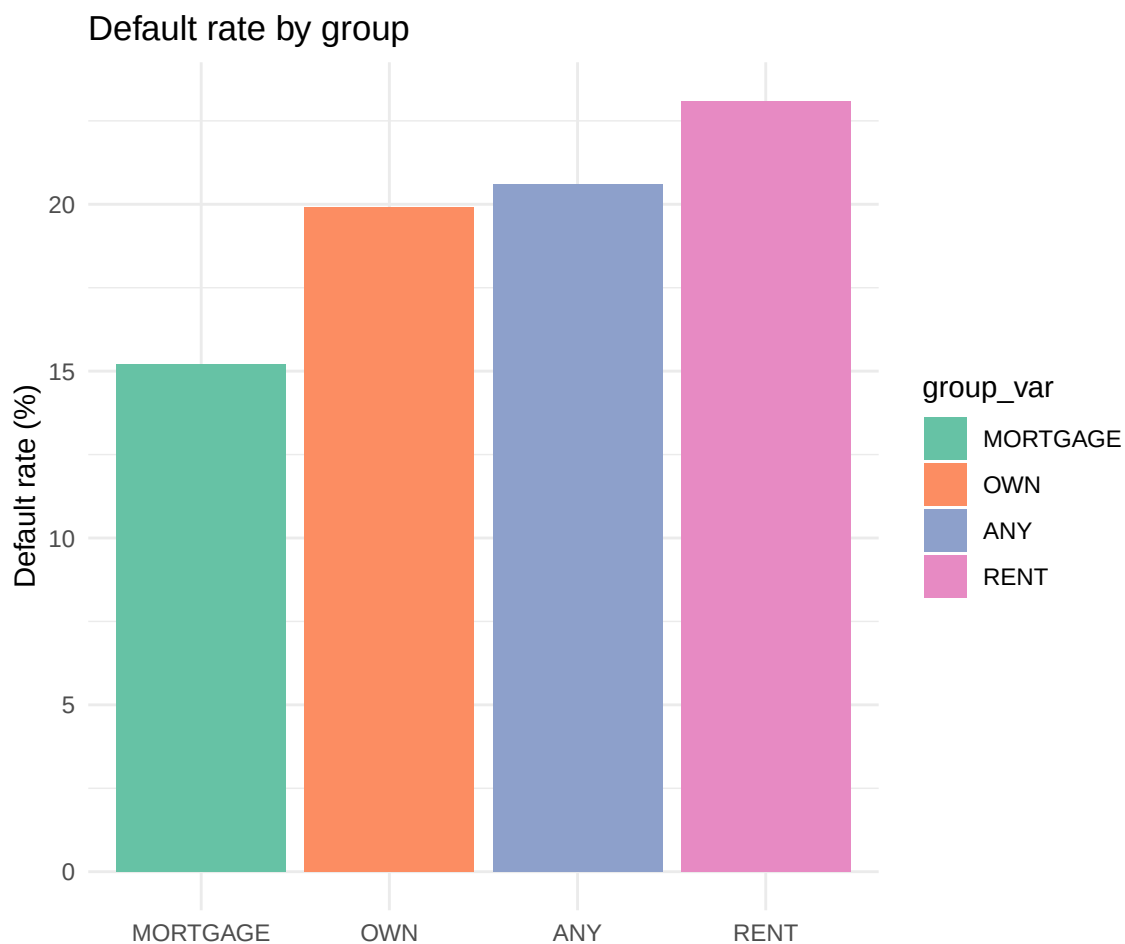


Figure 2.1: Home ownership and default rates.

Employment length does not appear to be a meaningful indicator of default on short-term loans. Employees with 10+ years have a slightly lower default rate, although the default rates between all employment length groups is very similar. The one notable exception is the group with a missing employment length, which defaults at 27.9%. This is possibly attributable to unemployed or self-employed individuals, who do not report their employment length.

Table 2.3: Income of missing employment observations

emp_missing	n	mean_income	median_income
FALSE	275792	79248.64	65000
TRUE	21655	49109.66	42700

3. State differences in short-term loan defaults

Now, I analyse differences in the defaults on short-term loans by state. There is significant variability in the average default rate across states. States such as Arkansas, Oklahoma, Louisiana, and Alabama have the highest default rates with more than 20%. The map plot confirms that there are interstate differences in short-term loan defaults. Specifically, Southern States have a higher propensity to default. The states with the lowest propensity to default are more Northern, coastal states, such as Oregon.

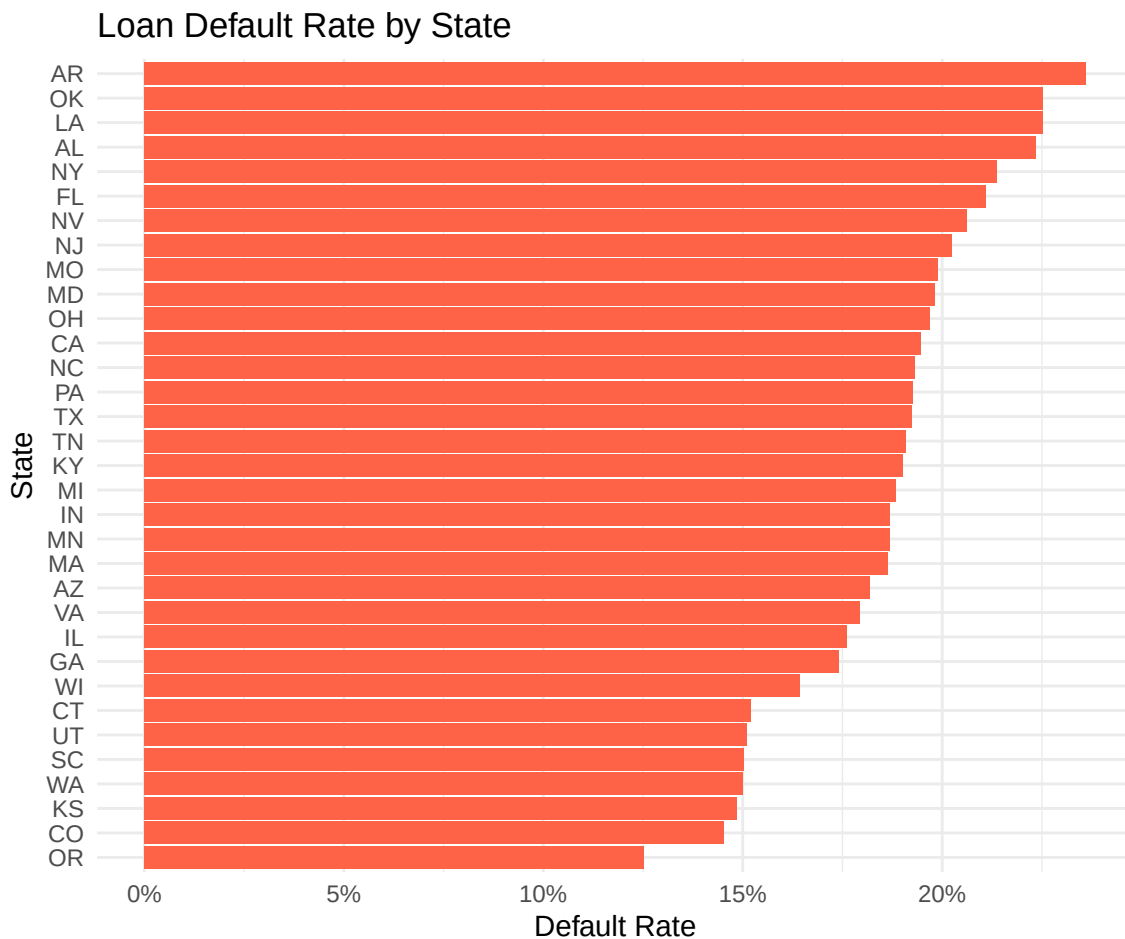


Figure 3.1: Default rate for American states.

Geographic Distribution of Loan Defaults

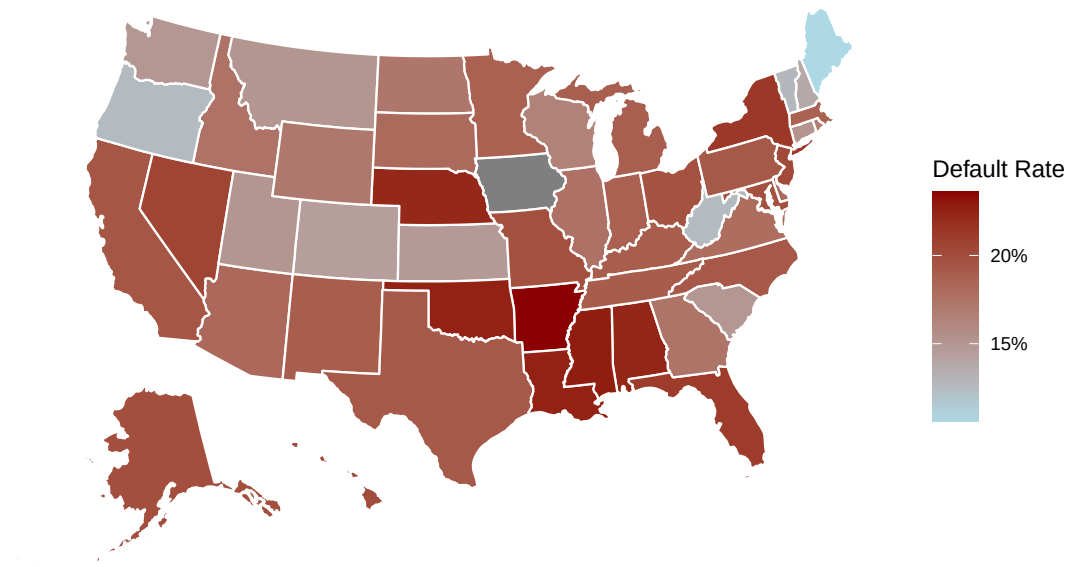


Figure 3.2: Default rate in the United States

4. What does grade predict?

The following table displays the relationship between average default rates and credit grades. As expected, the default rate rises steadily from Grade A to Grade G. This indicates the credit grading system tracks default risk well. The previous section highlighted two states of interest, Arkansas and Oregon, representing the top and bottom of the load default rate spectrum, respectively. Therefore, I plot the average default rate for each credit rate for both these states. I add Texas as it fell in the midrange of the state loan default rate.

Table 4.1: Default rate by credit grade

grade	n	default_rate
G	523	56.2

grade	n	default_rate
F	2141	48.0
E	9862	39.7
D	35625	32.5
C	83204	24.0
B	101540	15.2
A	64552	6.7

As expected, Arkansas generally has a higher default rate at each credit rating level, followed by Texas and Oregon. This indicates that the grade-default relationship is consistent in this three state sub-sample. The only exception is Grade G, which makes up a small amount of the observations, indicating that the prominence of Grade G in the plot may be unreliable.

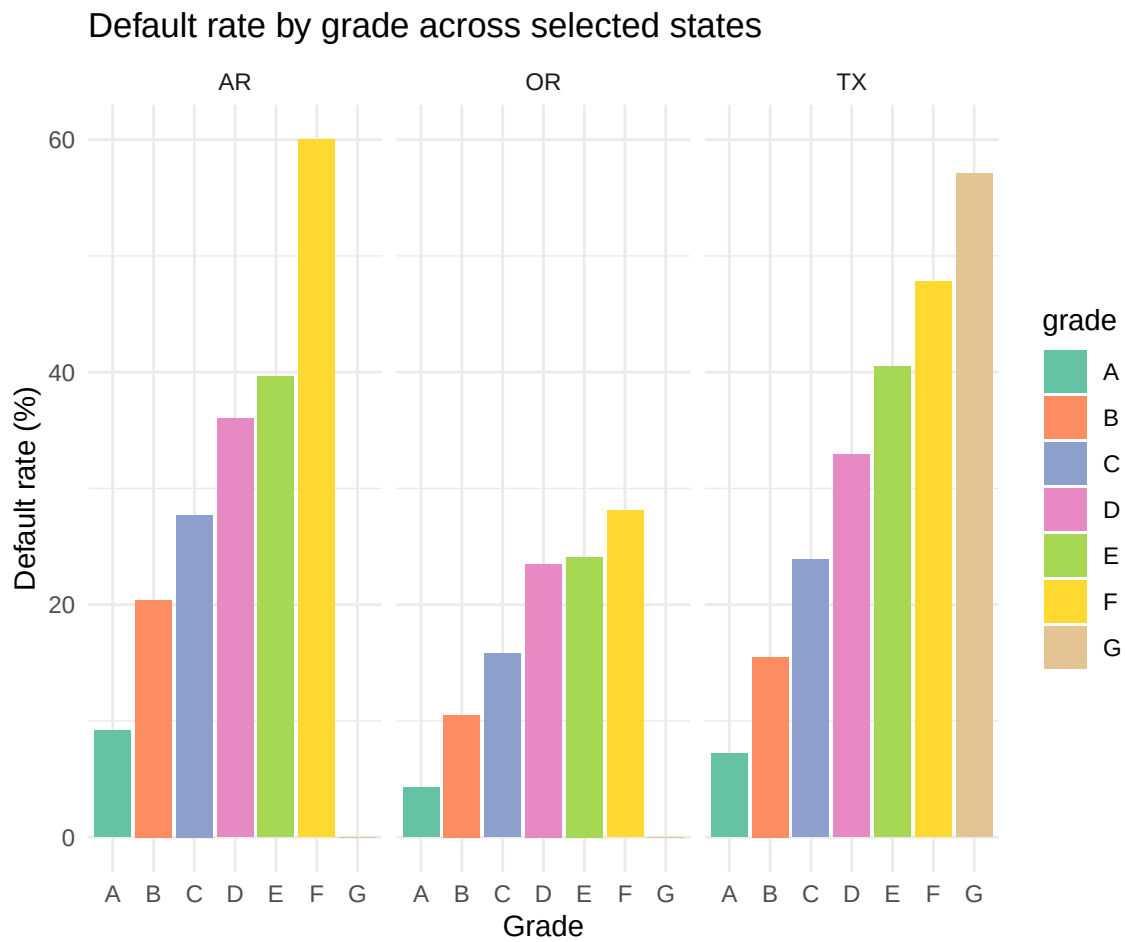


Figure 4.1: Default rate by grade across Arkansas, Oregon, and Texas.